

Let's get something out of the way first. Data is not clean. Raw data, the kind that analysts and programmers deal with on a day to day basis, is usually disorganized, poorly maintained and even erroneous. The primary objective behind the data analytical procedure however is to derive meaningful conclusions and results from the data sets. But if the system takes in garbage values, it will output garbage results. How do we deal with such complications?

Data preparation is regarded as an integral aspect of any data project that it commands its own team of programmers and workers. It's become an important step in the steps of data refining and preprocessing that precede the analytical steps.



Data Processing Steps

WHY DATA IS NEVER CLEAN

“Wouldn't better collection methods help in cutting down on preparation?”, you might ask. Sadly, the technology that exists for keeping and collecting data is never presented in a format fit for data analysis. Given the large number of different data types available, streamlining all the various types of data into a singular system can be a daunting task. It can be made even more challenging if there is no proper prior clean up.

Think about data types. Even though you may clean the columns that represent time in an orderly fashion, the system may not interpret it the same way as other numeric data types. And what about visual media – texts, videos, images, real time information? Much of the reason for why data preparation is necessary is because thoroughly checking the data not only helps remove kinks but also identify problems with the values.

In industries, sound data management is even more important as it keeps the wheels of the processes running with no delays. Erroneous results are usually due to poor data cleaning methods or the poor management of datasets, making them vulnerable to misinterpretations. Another purpose for data preparation is to convert contextual data to a format that the system